

Abstract—The growing integration of artificial intelligence across cloud, edge, and IoT infrastructures makes cybersecurity, resilience, and digital sovereignty an urgent strategic challenge for Europe. As computation and intelligence become distributed across a heterogeneous computing continuum, traditional perimeter-centric security models are no longer sufficient. This challenge is particularly acute given the simultaneous operationalisation of NIS2, the Cyber Resilience Act, the AI Act, and DORA, together with increasing concern over Europe’s structural dependencies in cloud ecosystems, hardware supply chains, and critical digital infrastructure. This paper addresses that gap by proposing a structured methodology for identifying, profiling, and prioritising cybersecurity capabilities for the European Cloud-Edge-IoT continuum. The methodology proceeds through three analytical stages: architectural decomposition into security layers and cross-cutting concerns, capability formulation through a seven-dimension profiling template, and priority clustering with phased transition matrices. Applying this methodology, the paper identifies ten strategic capabilities spanning hardware roots of trust, post-quantum cryptographic agility, lightweight edge security, federated interoperability, supply chain verification, cognitive security operations, pan-European network federation, confidential computing, resilient infrastructure, and proactive threat management. These capabilities are consolidated into three executive priority clusters — Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty — and analysed across a 2026–2040 transition horizon. The paper further advances the concept of autonomous, federated, and certified Security-by-Design as a strategic target for Europe, and derives implementation-oriented recommendations linking research, policy, industrial capability, and sovereignty objectives. It is relevant both as a methodological contribution to cybersecurity roadmapping and as a timely strategic framework for Europe’s AI-enabled, resilient, and sovereign digital future.

Index Terms—Cloud-Edge-IoT continuum, cyber-resilience, cybersecurity roadmap, digital sovereignty, European Union, federated security, Security-by-Design.

Cybersecurity and Cyber-Resilience for the European Cloud-Edge-IoT Computing Continuum: A Strategic Capability Roadmap

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I. INTRODUCTION

The ongoing evolution from centralised cloud infrastructures toward a distributed Cloud-Edge-IoT continuum is fundamentally reshaping the design, deployment, and operation of digital systems [42, 43, 44, 45]. By moving computation, storage, and intelligence closer to end devices, users, and physical processes, this paradigm enables low-latency services, real-time decision-making, operational autonomy, and scalable data-driven innovation [46, 47, 48, 49, 50, 51] across critical domains such as industry, mobility, energy, healthcare, telecommunications, and public services. However, this architectural shift also introduces a substantially more complex cybersecurity landscape.

Unlike centralised environments, the Cloud-Edge-IoT continuum is characterised by decentralisation, heterogeneity, physical exposure, and the coexistence of multiple administrative and trust domains [64, 65]. Security can no longer rely primarily on static perimeters, centralised visibility, or uniform infrastructure assumptions. Instead, defenders must secure systems that are geographically dispersed, operationally diverse, resource-constrained, and often exposed to both cyber and physical threats [53, 66]. As a result, traditional reactive and siloed security models are increasingly insufficient for addressing the scale, dynamism, and interdependence of the continuum [57, 67].

This challenge is particularly important in the European context. Europe has established a strong regulatory and normative foundation for trustworthy digital systems through instruments such as NIS2 [1], the Cyber Resilience Act [2], the AI Act [3], and the evolving EU cybersecurity certification framework [60, 61]. NIS2 establishes a unified legal framework across critical sectors and strengthens risk management, reporting, cooperation, and enforcement, though transposition has proven significantly delayed, with approximately 20 of 27 Member States having completed national implementation by early 2026 against a legal deadline of October 2024, and the Commission having issued reasoned opinions to 19 Member States in May 2025 [4]. The AI Act introduces a risk-based framework for trustworthy AI, with high-risk system obligations under Articles 9, 13, and 15 entering applicability from August 2026 under the default timetable, though the Digital Omnibus on AI (COM(2025) 836) proposes deferral to December 2027 for certain categories [5]. EUCC provides the first operational European cybersecurity certification mechanism for ICT products [6], while EUCS

for cloud services remains unadopted as of April 2026 — a structural policy impasse now likely to be resolved through the 2026 Cybersecurity Act revision (COM(2026) 13) rather than through the original ENISA candidate scheme process [7].

At the same time, Europe's position remains asymmetric. While it is comparatively strong in trust, privacy, certification, and governance, it still faces structural dependencies in cloud ecosystems, digital infrastructure, and semiconductor supply chains. The European Chips Act [8] explicitly links semiconductor capacity to technological sovereignty and reduced external dependency, with the EU targeting 20% of global semiconductor production by 2030 — though the European Court of Auditors' Special Report 12/2025 projects a more realistic trajectory of approximately 11.7%, and the cancellation of Intel's Magdeburg flagship fabrication facility in July 2025 has underscored the fragility of this ambition [9]. Recent analyses continue to highlight dependence on non-EU software and cloud providers, with Synergy Research Group reporting that AWS, Azure, and GCP collectively hold approximately 70% of the European cloud market as of mid-2025 [10], alongside broader concerns about strategic reliance on foreign actors and single service providers.

Against this background, the strategic goal of this paper is to support the transition from *reactive, siloed security* toward *autonomous, federated, and certified Security-by-Design* [56, 57, 58, 59] for the Cloud-Edge-IoT continuum. In this paper, *autonomous* refers to infrastructures capable of sustaining trustworthy and resilient operation under dynamic and adverse conditions; *federated* refers to interoperable security and trust mechanisms spanning multiple providers, sectors, and jurisdictions, like SSO systems; and *certified Security-by-Design* refers to embedding assurance, compliance, and trust requirements throughout the lifecycle of systems, components, and services rather than applying them retrospectively.

The main argument of the paper is that Europe's comparative advantage in trust, privacy, governance, and certification can become strategically meaningful only if it is translated into deployable cybersecurity capabilities and coherent technical priorities. Therefore, this paper approaches cybersecurity not merely as a narrow technical protection problem, but as a combined challenge of architecture, operational resilience, governance, interoperability, and sovereignty [60, 61, 62, 63]. On this basis, the paper identifies and structures the cybersecurity priorities that are most critical for building a secure, resilient, and sovereign European Cloud-Edge-IoT ecosystem.

The paper makes the following contributions. It frames cy-

bersecurity in the Cloud–Edge–IoT continuum as a multi-layer challenge spanning technical security, operational resilience, governance, interoperability, and digital sovereignty [52, 53, 54, 55]. It advances a strategic transition from reactive, siloed protection models toward autonomous, federated, and certified Security-by-Design [56, 57, 58, 59]. It proposes a three-stage methodology — architectural decomposition, seven-dimension capability profiling, and priority clustering with transition matrices — that connects technical cybersecurity capabilities with broader European strategic concerns including sovereignty, trust, and regulatory alignment. It identifies and profiles ten strategic capabilities and consolidates them into three priority clusters analysed across a 2026–2040 horizon. And it provides a prioritisation-oriented foundation that can support research and innovation planning, policy alignment, and deployment strategy in Europe.

The remainder of this paper is organised as follows. Section 2 presents the background, state of the art, and key driving factors, with particular attention to the European position. Section 3 introduces the methodology. Section 4 presents the architectural layers and corresponding focus areas. Section 5 analyses the ten identified capabilities through the seven-dimension profiling template. Section 6 synthesises these capabilities into priority clusters and transition matrices. Section 7 presents strategic recommendations. Section 8 discusses validation, limitations, and methodological positioning. Section 9 concludes.

II. BACKGROUND, STATE OF THE ART, AND DRIVING FACTORS

The transition from centralised cloud computing toward a distributed Cloud–Edge–IoT continuum is reshaping the cybersecurity landscape. While this paradigm enables low-latency services, localised intelligence, real-time analytics, and increased operational autonomy, it also introduces a fundamentally more complex security environment. In contrast to the relatively controlled conditions of centralised hyperscale cloud infrastructures, the continuum is characterised by large numbers of heterogeneous, geographically dispersed, and resource-constrained nodes operating across multiple administrative and trust domains [64, 65]. As a result, the attack surface expands significantly, creating infrastructures that are, in practice, too numerous, too diverse, and too dispersed to be secured through traditional perimeter-oriented approaches alone [57, 67].

A central challenge stems from decentralisation itself. Security models developed for centralised cloud environments assume relatively stable boundaries, consolidated monitoring, and strong physical and logical control over infrastructure. These assumptions do not hold in edge and IoT deployments, where services, workloads, and data increasingly move closer to end users, devices, and operational environments. This decentralisation complicates visibility, orchestration, patching, identity management, and incident response, while also increasing the likelihood of inconsistent security postures across the continuum.

A second major challenge is the physical exposure of edge infrastructure. Unlike hyperscaler data centres, edge nodes

are often deployed in the field — in factories, smart-city installations, transport systems, telecom sites, or remote industrial environments. These deployments rarely benefit from the same degree of physical protection, controlled access, or environmental safeguards as centralised facilities. Consequently, they are more vulnerable to device tampering, hardware manipulation, fault injection, side-channel attacks, and supply-chain compromise [53, 66]. This makes hardware-rooted trust, secure boot, attestation, and tamper-aware system design increasingly important as baseline requirements rather than optional enhancements.

Simultaneously, the cybersecurity landscape is being shaped by an unprecedented regulatory push. In Europe, the operationalisation of NIS2 [1], the Cyber Resilience Act [2], the AI Act [3], and the Digital Operational Resilience Act (DORA) [11] is creating immediate pressure for organisations to demonstrate security, resilience, accountability, and compliance across distributed and interconnected infrastructures. The challenge is not only legal alignment, but technical translation: regulatory obligations must be converted into deployable security architectures, assurance mechanisms, lifecycle controls, and governance models that can function across a fragmented ecosystem of cloud, edge, and IoT stakeholders. The simultaneity of this regulatory activation — with CRA enforcement beginning September 2026 and full application in December 2027, AI Act high-risk provisions from August 2026, DORA fully applied from January 2025, and the Data Act [12] operational from September 2025 — creates a compliance coordination challenge that disproportionately burdens SMEs and smaller Member States.

These technical and regulatory pressures are closely linked to the broader strategic objective of digital sovereignty. As Europe expands its digital infrastructure and data-driven services, systemic dependence on non-EU providers for key hardware, platforms, and security-critical components introduces both geopolitical and operational risk. Extra-territorial data access concerns — underscored by Microsoft France’s June 2025 confirmation before the French Senate that it cannot guarantee EU data will never be accessed under the US CLOUD Act [13] — foreign technology dependencies, and vendor lock-in can undermine Europe’s capacity to guarantee autonomy, trust, and resilience in critical digital infrastructures.

A. Europe in Relation to the Current State of the Art

In relation to the global state of the art, Europe holds a distinctive position. Its comparative advantage lies less in hyperscale platform dominance and more in its leadership in trust, privacy, governance, and certification [60, 61, 62]. Europe has established a world-leading regulatory and normative foundation through instruments such as the GDPR [14], NIS2, the CRA, and the AI Act. Together, these frameworks embed privacy-by-design, trust-by-design, and accountability into the legal and operational fabric of digital systems. In parallel, European certification initiatives — notably EUCC, which became operational on 27 February 2025 under Commission Implementing Regulation (EU) 2024/482 [6] — provide a rigorous basis for assurance and trusted evaluation. The European Cybersecurity Competence Centre (ECCC) [15], the

national coordination centres network, and the Horizon Europe funding ecosystem create institutional support for long-term research, innovation, and deployment. Initiatives such as the Simpl middleware programme [16] and the Common European Data Spaces further demonstrate Europe’s ambition to build interoperable, federated, and trustworthy digital infrastructures.

However, these strengths coexist with significant structural gaps. Europe remains critically dependent on non-EU secure silicon, including trusted platform modules, secure enclaves, and other hardware roots of trust. While EU-headquartered vendors — Infineon, NXP, STMicroelectronics — design the dominant TPM products and have achieved EUCC ‘High’ certification for multiple secure element families (STMicroelectronics ST54J/K certified by ANSSI on 1 April 2025; NXP SN300/JCOP in July–August 2025; Infineon IFX_CCI in December 2025 [6]), fabrication depends on non-EU foundries (primarily TSMC) and the designs remain proprietary, preventing sovereign auditability at the silicon level. Intel, AMD, and ARM control the entire operational trusted execution environment market. Threat intelligence and operational cyber defence capabilities remain fragmented across national and sectoral boundaries, with no unified pan-European cyber situational awareness capability at scale. The enterprise SIEM market is dominated by US platforms — Splunk/Cisco, Microsoft Sentinel, and Palo Alto Networks (which acquired IBM QRadar SaaS in August 2024) — with no EU-headquartered vendor in the Gartner Leaders quadrant as of October 2025 [17]. Europe also lacks production-scale PQC-capable hardware tailored for edge and IoT environments, though the NIS Cooperation Group’s “Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography” (v1.1, 23 June 2025) [18] now provides a binding Member State migration framework with milestones at 2026, 2030, and 2035.

Taken together, this position suggests that Europe is not starting from weakness, but from asymmetry. It is strong in governance, trust, assurance, and regulatory design, yet weaker in industrial scale, unified operationalisation, and sovereign control of critical enabling technologies. This gap between normative leadership and deployable capability — what we term the *ambition–implementation gap* — defines the central challenge for the next phase of European cybersecurity strategy in the Cloud–Edge–IoT continuum.

III. SYSTEM MODEL OF THE CLOUD-EDGE-IOT ECOSYSTEM

These are

Vertical Stack:

- **Layer 1: Hardware-Centric Security**
- **Layer 2: Software-Centric Security (Middleware/OS)**
- **Layer 3: Application-Centric Security (Apps/Services)**
- **Layer 4: User & Orchestration-Centric Security**
- **Layer 5: Data-Centric Security**
- **Layer 6: Network Infrastructure & Perimeter-Centric Security**

The Transversal Verticals (Cross-Layers):

- **Cross-Layer 1: Automated Security Processes & Cyber-Resilient AI Lifecycle**

- **Cross-Layer 2: Dynamic Risk Management & Operational Cyber Resilience**
- **Cross-Layer 3: Interoperable Security Mechanisms & Open-Source Verifiability**
- **Cross-Layer 4: Compliance-by-Design, Agile Governance & Automated Assurance**

IV. METHODOLOGY

The methodology presented in this section is designed to transform a broad and heterogeneous strategic landscape — spanning hardware security, post-quantum cryptography, AI-driven defence, federated governance, and supply chain integrity across the European cloud–edge–IoT computing continuum — into a structured roadmap for research, innovation, policy, deployment, and ecosystem development. The analytical process proceeds through three staged phases — architectural decomposition, capability formulation and profiling, and priority clustering with phased progression — followed by an iterative validation step.

Two foundational commitments shape the methodology. First, it is intended to support strategic roadmap construction, not deterministic forecasting. In complex socio-technical ecosystems, progress is non-linear and depends on interacting technical, regulatory, operational, and market developments whose timing and interdependencies cannot be predicted with precision. Second, the methodology is designed to connect technical cybersecurity capabilities with broader European strategic concerns — resilience, interoperability, trust, and what we term *Sovereignty Level*: the degree to which critical capabilities, control points, and dependencies remain under credible European influence, governance, or operational control. Sovereignty Level is employed throughout the methodology as an interpretive strategic lens rather than as a rigid scoring model; it guides assessment of whether progress in cybersecurity and cyber-resilience strengthens Europe’s strategic control, trusted autonomy, and reduced dependency in critical parts of the computing continuum.

The methodological process is structured into four interconnected steps:

- **Stage 1. Core Dimensions and Focus Areas Identification:** which establishes the analytical structure
- **Stage 2. Capability Identification and Profiling:** which translates problem domains into strategic capabilities evaluated through a consistent analytical template
- **Stage 3. Priority Clustering and Transition Matrix:** which groups capabilities into broader strategic priorities and positions them across phased pathways
- **Validation step:** drawing on expert review, consultation, and iterative refinement.

A. Stage 1: Core Dimensions and Focus Area Identification

The first stage provides the analytical scope and structure of the roadmap by decomposing the cloud–edge–IoT computing continuum into a set of architectural vertical and cross-cutting layers. These dimensions are treated analytically distinct but operationally interdependent: a vulnerability or governance gap at one layer typically has consequences that propagate

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Fig. 1. Figure 1: Security Framework with separated security verticals in dedicated boxes

to adjacent layers, and effective security requires coordinated action across the full vertical and horizontal extent of the architecture.

The purpose of this stage is to answer three questions. Where are the major security and resilience problem domains within the distributed computing continuum? At what architectural layers or cross-layers do these problems emerge? And which focus areas are sufficiently strategic, cross-cutting, or underdeveloped to justify dedicated roadmap attention?

Focus areas are derived from four complementary sources. The first is the architectural characteristics of distributed cloud-edge-IoT systems themselves: the heterogeneity of execution environments spanning HPC, cloud, edge, and device tiers creates structurally distinct security requirements at each level of the stack. The second is the set of recurring cybersecurity and cyber-resilience challenges observed across these tiers — hardware integrity, software supply chain verification, workload isolation, identity federation, data confidentiality, and network perimeter control. The third comprises cross-cutting strategic concerns that cannot be localised to a single layer: sovereignty, trust, interoperability, resilience, assurance, and regulatory alignment. The fourth is the practical need to connect technical architecture with operational and governance realities — recognising that a capability with no credible deployment pathway or regulatory anchor is unlikely to achieve ecosystem-wide impact regardless of its technical merit.

The output of Stage 1 is not a priority list but a structured map of the problem space, organised into six Security Layers (the vertical stack) and four Transversal Cross-Layers (the horizontal enablers), as detailed in Section 4. Within this structured map, some focus areas acquire greater strategic importance because they influence Europe's future Sovereignty Level — particularly where dependencies on external providers, fabrication capacity, or assurance mechanisms are structurally significant. Stage 1 identifies these sovereignty-sensitive focus areas explicitly so that subsequent stages can track their evolution as a strategic concern alongside purely technical maturity.

B. Stage 2: Capability Identification and Profiling

Once the architectural map and its constituent focus areas are defined, the methodology moves from problem framing to capability formulation. A capability, in the context of this roadmap, is understood as a strategic ability that the European ecosystem should be able to design, deploy, govern, standardise, assure, coordinate, or scale in order to achieve trustworthy, resilient, and sovereign operation of the computing continuum. Capabilities are not limited to technical controls; they may also encompass governance mechanisms, interoperability instruments, assurance functions, ecosystem coordination capacities, and deployment enablers.

Capabilities are derived by translating each focus area identified in Stage 1 into actionable strategic needs, guided

by a consistent question: what must Europe be able to do — technically, institutionally, and industrially — to address this focus area at the scale and with the assurance level that the regulatory and threat environment demands? The ten capabilities identified through this process (detailed in Section 5) span the full range from hardware roots of trust and post-quantum cryptographic agility to federated security interoperability and proactive cyber threat management.

1) The Seven-Dimension Capability Profile: Each capability is assessed through a seven-dimension analytical template applied identically across all priorities. This template ensures that every capability is characterised not only from a technical perspective, but also in terms of its regulatory alignment, investment pathway, failure modes, and contribution to European strategic autonomy. The consistency of the template enables structured cross-priority comparison — a feature essential both for funding prioritisation and for identifying shared dependencies and bottlenecks that span multiple capabilities.

Dimension 1: Baseline and Target Objective. This dimension grounds the capability in reality by describing the present condition of the European ecosystem, the intended future condition, and the strategic change the capability is meant to achieve. The baseline captures current industrial capacity, the degree of reliance on non-European vendors, technical and economic barriers to adoption, and the state of standardisation and certification. The target defines measurable end-states — quantified market share objectives, certification levels, migration coverage metrics, or deployment thresholds.

Dimension 2: Primary Drivers. This dimension articulates the external pressures that make the capability urgent and non-deferrable. It distinguishes between threat drivers (adversarial capabilities or attack vectors that are already active or imminently scalable) and regulatory drivers (specific EU legal obligations with enforcement timelines, citing the precise directive or regulation and its relevant provisions).

Dimension 3: High-Impact Activities. This dimension identifies three to five concrete, actionable research and innovation activities expected to yield the greatest strategic return on investment. Each activity is tagged with its primary target phase (Phase 1, Phase 2, or Phase 3).

Dimension 4: Catalysts and Enablers. This dimension identifies the concrete mechanisms, funding vehicles, and regulatory standardisations required to accelerate the capability from baseline toward target. Three categories are distinguished: R&I and funding alignment (Horizon Europe clusters, Digital Europe Programme, ECCC Strategic Agenda), regulatory and standardisation (CEN/CENELEC, ETSI, ISO standards), and ecosystem enablers (pilot infrastructures, industrial partnerships, SME capacity-building).

Dimension 5: Disruptors and Systemic Risks. This dimension identifies risks that could render planned deployments obsolete or unachievable: technical cliffs (defensive capabilities rendered ineffective by adversarial breakthroughs), supply

chain and market risks (fabrication bottlenecks, vendor resistance, monopolisation), and geopolitical risks (export controls, extra-territorial jurisdiction conflicts).

Dimension 6: Failsafe Mechanisms and Resilience. This dimension specifies engineering fallbacks if a capability's primary mechanism is compromised, delayed, or only partially realised: hybrid transitional schemes, graceful degradation modes, autonomous isolation protocols, and implementation diversity mandates.

Dimension 7: Transversal Meta-Layer (Policy Alignment). This dimension connects the capability to the EU's overarching strategic autonomy agenda across three axes: sovereignty and autonomy, trust and ethics, and sustainable security.

C. Stage 3: Priority Clustering and Transition Matrix

Not every capability can or should be treated as a standalone roadmap priority. Capabilities that share architectural proximity, common enabling conditions, similar policy needs, overlapping ecosystem actors, or convergent strategic outcomes benefit from coordinated treatment. Stage 3 therefore groups the ten capabilities into three broader Priority Clusters:

- **Priority Cluster 1: Sovereign Trusted Infrastructure** — the hardware, cryptographic, and resilience foundations that anchor trust across the continuum.
- **Priority Cluster 2: Cognitive Security and Autonomous Cyber Defence** — the AI-driven operational layer that scales defence beyond human-paced response.
- **Priority Cluster 3: Federated Operational Sovereignty** — the interoperability, supply-chain, and federation substrate that enables a unified European trust fabric.

Each cluster is analysed through a three-phase transition matrix:

- **Phase 1: Near-Term Implementation and Pilot Actions (2026–2030)** — deploying near-ready solutions, enforcing enacted regulation, and closing governance gaps before technical work scales.
- **Phase 2: Mid-Term Scale-Up, Industrial R&D and Standardisation (2030–2035)** — industrialising sovereign capabilities, federating architectures, and building the workforce and institutional infrastructure the technical work requires.
- **Phase 3: Long-Term Strategic Autonomy, Frontier R&I and Leadership (2035–2040)** — achieving full operational maturity, pioneering frontier research, and resolving structural doctrine and oversight questions.

1) Analytical Perspectives: For each priority cluster and phase, the assessment considers four analytical perspectives: *R&I maturity* (technical readiness and industrialisation feasibility), *policy, regulation, and standards* (legislative enactment, enforcement credibility, standards availability, conformity assessment capacity), *operational deployment and adoption* (movement beyond laboratory environments into operational use), and *market uptake and strategic impact* (commercial traction, European provider competitiveness, measurable Sovereignty Level improvement).

2) Transition Conditions: This roadmap uses *transition conditions* rather than hard gates between phases. Transition conditions are evidence-informed progression markers indicating when a priority cluster appears sufficiently prepared to advance. They are framed as analytical signals of increasing readiness, coherence, and ecosystem support — not binary thresholds or deterministic requirements. This design choice reflects the non-linear nature of progress in complex socio-technical ecosystems, where adversarial breakthroughs, regulatory delays, fabrication bottlenecks, and geopolitical disruptions can alter timelines unpredictably.

A transition condition is a short statement indicating that a technical basis has become sufficiently mature for the next stage of deployment; a policy or standardisation basis has emerged that reduces adoption uncertainty; a deployment pathway has been validated through pilots; a strategic ecosystem condition has materialised; or a Sovereignty Level improvement has become plausible through reduced dependency or increased European capacity.

Transition conditions are not promises, deterministic predictions, or precise quantitative thresholds unless evidence genuinely supports them. They should be interpreted as structured indicators of directional readiness — useful for strategic planning and coordination, but subject to reassessment as the ecosystem evolves.

V. CORE DIMENSIONS AND FOCUS AREAS

The roadmap is structured across a cohesive architectural framework spanning four continuum environments — HPC, Cloud, Edge, and Devices. The capabilities are organised into six Security Layers (the vertical stack) and four Transversal Cross-Layers (the horizontal enablers). This section describes each dimension and its constituent focus areas.

A. Security Layers (Vertical Stack)

Layer 1: Hardware-Centric Security. This layer addresses the physical and silicon-level trust foundations of the computing continuum. It encompasses two focus areas. *Layer 1.1 (Hardware Roots of Trust)* concerns cryptographically verifiable supply chain trust ensuring physical device integrity from fabrication to deployment, with a strategic focus on driving European open-hardware designs (RISC-V) toward EUCC 'High' assurance certification [6]. *Layer 1.2 (Physical-Layer Threat Detection and Fingerprinting)* exploits intrinsic wireless channel characteristics — Channel State Information, Carrier Frequency Offset, and RSSI — to establish real-time device authentication and anomaly detection at the lowest communication layer, complementing upper-layer cryptographic protections.

Layer 2: Software-Centric Security (Middleware/OS). This layer secures the software substrate on which continuum workloads execute. *Layer 2.1 (OS and Middleware Hardening)* addresses the security of hypervisors, container runtimes, and distributed operating systems, with emphasis on sovereign, lightweight Linux distributions optimised for edge resource constraints and reduced attack surfaces. *Layer 2.2 (Secure OTA Patch and Update Management)* recognises that edge nodes

deployed in the field — smart grids, O-RAN radios, industrial gateways — require cryptographically verified, failure-resilient over-the-air update pipelines to maintain security posture without physical access.

Layer 3: Application-Centric Security (Apps/Services). This layer protects the workloads and service interfaces that execute across the continuum. *Layer 3.1 (Workload Security)* secures containerised applications and serverless functions through hardened service meshes and sidecar proxy architectures providing mutual TLS and granular traffic inspection. *Layer 3.2 (In-Band Network Telemetry and Deep Observability)* embeds smart telemetry directly into data packets traversing containerised workloads, enabling real-time security monitoring without dedicated out-of-band infrastructure. *Layer 3.3 (API Security Gateway and Policy Enforcement)* provides a unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection as containerised AI workloads increasingly communicate through REST/gRPC APIs.

Layer 4: User and Orchestration-Centric Security. This layer governs identity, access, and policy enforcement across the multi-provider continuum. *Layer 4.1 (Zero-Trust Orchestration)* implements Policy Enforcement Point / Policy Decision Point architectures enabling continuous authorisation and dynamic access controls adapted to request context, aligned with NIST SP 800-207 [19]. *Layer 4.2 (Federated Identity and Access Management)* utilises Self-Sovereign Identity principles for cryptographically secure, decentralised authentication of machines, users, and automated workloads. *Layer 4.3 (Privileged Access Management and Just-In-Time Access)* addresses the lateral movement vector behind major APT campaigns by replacing over-provisioned standing privileges with time-bounded, contextually justified access grants.

Layer 5: Data-Centric Security. This layer provides end-to-end data protection across the continuum lifecycle. *Layer 5.1 (Confidential Computing)* delivers data-in-use security through Trusted Execution Environments spanning the Cloud-to-IoT data path. *Layer 5.2 (Post-Quantum Cryptography)* addresses the migration to quantum-resistant algorithms — ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SLH-DSA (FIPS 205), finalised by NIST in August 2024 [20][21][22] — for resource-constrained devices to neutralise “Harvest Now, Decrypt Later” threats. *Layer 5.3 (Signalling Data Security)* embeds trust directly into network signalling architecture. *Layer 5.4 (Distributed Reputation and Trust Networking)* implements real-time reputation systems built on distributed ledger technology. *Layer 5.5 (Quantum Key Distribution for Critical Links)* provides information-theoretically secure key exchange for high-value point-to-point links. *Layer 5.6 (Secure Multi-Party Computation)* enables multiple mutually distrusting parties to jointly compute functions over combined private inputs without revealing those inputs.

Layer 6: Network Infrastructure and Perimeter-Centric Security. This layer secures the communication fabric of the continuum. *Layer 6.1 (Telco-cloud and O-RAN Security)* hardens distributed 5G/6G cores and RAN components in alignment with the SNS JU SRIA [23], ensuring sovereign control over network slice isolation. *Layer 6.2 (Physical*

Resilience and Safety) adapts IT/OT gateway security for high-availability edge environments to prevent digital breaches from causing kinetic damage. *Layer 6.3 (Edge-Native Autonomous Guardians)* hosts lightweight, localised security control functions directly at far-edge gateways or within Open-RAN Radio Intelligent Controllers. *Layer 6.4 (Non-Terrestrial Network and Satellite Security)* defines security requirements for 6G NTN integration aligned with 3GPP NTN specifications [24] and the EU Space Programme.

B. Transversal Cross-Layers (Horizontal Enablers)

Cross-Layer 1: Automated Security Processes and Cyber-Resilient AI Lifecycle. This cross-layer spans the full lifecycle of software and AI component security. *CL1.1 (Continuous DevSecOps)* embeds AI-driven code analysis and automated vulnerability remediation at the earliest stages of the software lifecycle. *CL1.2 (Model Integrity and DataBOMs)* secures AI training pipelines against data poisoning and unauthorised model manipulation. *CL1.3 (Dynamic Supply Chain Verification and Runtime SBOMs)* implements dynamic Software Bills of Materials and cryptographic proof of execution to continuously verify the integrity of containerised microservices and AI components as they migrate across the multi-provider continuum. *CL1.4 (Automated Adversarial Testing and Continuous Purple Teaming)* provides the adversarial counterpart to DevSecOps: continuous automated penetration testing, fuzzing, and purple-team automation against live staging environments.

Cross-Layer 2: Dynamic Risk Management and Operational Cyber Resilience. This cross-layer governs real-time threat detection, analytics, and operational response. *CL2.1 (Automated SOC and Threat Intelligence)* enables SOC federation across Member States using legally protected platforms to overcome institutional trust barriers [25]. *CL2.2 (High-Precision Telemetry and Security Signalling)* implements smart telemetry to feed real-time threat intelligence without overwhelming network bandwidth. *CL2.3 (Internet of Behaviour and Evidence Sharing)* extracts and shares behavioural patterns across the infrastructure. *CL2.4 (AI-Native Analytics Integration via NWDAF and LLMSecOps)* routes cybersecurity telemetry through centralised analytics hubs using 5G NWDAF architecture [26] and large language model-based security operations. *CL2.5 (Cyber Threat Intelligence Lifecycle Management)* governs the full CTI lifecycle as a prerequisite for the federated SOC vision. *CL2.6 (Deception Technology and Active Defence)* deploys decoy assets — honeypots, honeynets, credential canaries, fake API endpoints — that generate high-fidelity, low-noise alerts upon adversary interaction.

Cross-Layer 3: Interoperable Security Mechanisms and Open-Source Verifiability. This cross-layer enables secure interoperation across heterogeneous providers and jurisdictions. *CL3.1 (Interoperability and Simpl Integration)* mandates integration into the Simpl open-source middleware [16] to unlock Common European Data Spaces. *CL3.2 (Neutral Host and Multi-Tenant Isolation)* enforces Zero-Trust edge operations across different network partitions under infrastructure sharing models. *CL3.3 (Pan-European NaaS Federation*

and *Cross-Border Edge Roaming*) establishes standardised interfaces leveraging CAMARA Network APIs [27] for a unified Network-as-a-Service platform. *CL3.4 (Semantic Security Interoperability)* ensures participating organisations share common ontologies for security events, asset taxonomies, and risk classifications. *CL3.5 (Cyber Range and Digital Twin Federation)* federates national cyber ranges into shared EU-wide validation infrastructure aligned with the Cyber Solidarity Act.

Cross-Layer 4: Compliance-by-Design, Agile Governance and Automated Assurance. This cross-layer embeds regulatory compliance into technical architecture. *CL4.1 (Post-Quantum Readiness and Zero-Trust Architectures)* embeds continuous, AI-driven Zero-Trust verification natively into cloud-edge infrastructure. *CL4.2 (Automated Certification)* replaces point-in-time audits with real-time certification tracking for cloud services (EUCS, when adopted) and edge products (EUCC) [6] throughout their lifecycle. *CL4.3 (Multicloud Key Management as a Service)* manages cryptographic keys across dynamic multi-provider computing continuum operations. *CL4.4 (Privacy-Preserving Federated Computation)* enables cross-jurisdictional AI training and inference over distributed private datasets. *CL4.6 (Cyber Risk Quantification and Board-Level Reporting)* translates technical security posture into financial and operational risk language for decision-makers. *CL4.7 (AI Trustworthiness and Algorithmic Accountability)* addresses the trustworthiness of AI systems that themselves make autonomous security decisions, in compliance with AI Act Article 9 requirements [3].

VI. CAPABILITY PROFILES

The ten strategic capabilities identified in this roadmap are analysed through the seven-dimension profiling template described in Section 3.2.1. This section presents each capability profile in full.

A. P1: Hardware Roots of Trust and EUCC ‘High’ Certification

Baseline and Target. The current European landscape reflects a structurally dependent but industrially active hardware security ecosystem. EU-headquartered vendors — Infineon, NXP, STMicroelectronics — design the dominant secure element products deployed globally, and all three achieved EUCC ‘High’ certification for multiple product families in 2025: STMicroelectronics ST54J/K (ANSSI, 1 April 2025), NXP SN300/JCOP (July–August 2025), and Infineon IFX_CCI G12 (December 2025) [6]. However, fabrication depends on non-EU foundries (primarily TSMC), and designs remain proprietary, preventing sovereign auditability at the silicon level. No EU-origin RISC-V processor has received EUCC or Common Criteria certification at any assurance level as of April 2026. Intel, AMD, and ARM control the entire operational TEE market. The target is ubiquitous deployment of EU-designed, open-source secure elements on EU-controlled fabrication capacity, certified at EUCC ‘High’, providing a verifiable trust foundation for the entire continuum. By 2040, at least 60% of critical infrastructure edge nodes should utilise EU-designed, EUCC ‘High’ certified secure elements.

Primary Drivers. *Threat:* state-sponsored exploitation of JTAG debug interfaces and side-channel attacks on field-deployed edge nodes; supply chain insertion of hardware backdoors. *Regulatory:* CRA Articles 10–11 imposing hardware security and lifecycle management requirements with enforcement from December 2027 [2]; NIS2 Article 21(2)(e) mandating cryptographic policies for essential entities [1]; AI Act Articles 9 and 15 requiring hardware integrity guarantees for high-risk AI systems [3].

High-Impact Activities. (Phase 1) Fund 3–5 RISC-V secure element pilots under Horizon Europe Cluster 3 targeting EUCC ‘High’ pre-evaluation; submit OpenTitan [28] and Keystone [29] prototypes for certification assessment. (Phase 2) Bring the first EUCC ‘High’ certified EU-manufactured RISC-V secure elements into production via EU Chips Act pilot lines, including NanoIC (imec, inaugurated February 2026) [8] and FAMES (CEA-Leti). (Phase 3) Establish EU-designed RISC-V secure elements as the hardware default across all new EU ICT products.

Catalysts and Enablers. EU Chips Act pilot fabrication lines [8]; ECCC Strategic Agenda [15] funding for open-hardware security; CEN/CENELEC JTC 13 hardware security standards; the DARE SGA1 RISC-V initiative (EuroHPC, launched March 2025, €240M) [30]; Giesecke+Devrient’s OpenTitan CC certification path.

Disruptors and Systemic Risks. EU Chips Act fabrication capacity shortfall — ECA Special Report 12/2025 projects only ~11.7% global semiconductor share by 2030, against the 20% target [9]. Intel Magdeburg cancellation (July 2025) removed the largest EU-based advanced node commitment. TSMC’s Dresden ESMC joint venture (28/22nm, production expected late 2027) does not address the leading-edge nodes required for advanced secure elements. Side-channel attack breakthroughs rendering current enclave architectures obsolete.

Failsafe Mechanisms. Design all pilot architectures with explicit TEE abstraction layers to avoid lock-in to Intel TDX, AMD SEV-SNP, or ARM TrustZone during the Phase 1 dependency period. Deploy software-based isolation (TEEs) as a secondary layer if specific hardware enclaves are compromised. Implement autonomous node quarantine via eBPF-based network policy enforcement if attestation fails.

Transversal Meta-Layer. *Sovereignty:* eliminates proprietary hardware dependencies and enables sovereign silicon auditability — the current dependency level is effectively 100% for TEE infrastructure and ~100% for fabrication. *Trust:* EUCC ‘High’ provides the highest assurance level under the EU cybersecurity certification framework, aligned with Common Criteria EAL4+ and AVA_VAN.5. *Sustainability:* energy-efficient secure silicon designs for far-edge deployment reduce the carbon footprint of cryptographic operations.

B. P2: Post-Quantum Cryptographic Agility for Edge/IoT

Baseline and Target. Current edge and IoT environments depend almost entirely on classical cryptographic algorithms (RSA, ECDH, ECDSA) vulnerable to future quantum computers. Cryptographic asset inventories are largely absent, and

migration readiness is weak outside government and financial sectors. The NIS Cooperation Group's PQC roadmap (v1.1, June 2025) [18] establishes a binding EU migration framework: national roadmaps and cryptographic inventories by December 2026; high-risk critical infrastructure transitioned by December 2030; full completion by December 2035. NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) were finalised in August 2024 [20][21][22]; FIPS 206 (FN-DSA/Falcon) remains pre-publication as of April 2026; HQC was selected as a backup KEM in March 2025 [31]. The target is a quantum-native continuum with full cryptographic agility enabling algorithm updates without hardware replacement.

Primary Drivers. *Threat:* the Q-Day horizon (current expert consensus: 10–20 years [32], with significant lower-bound uncertainty) and active “Harvest Now, Decrypt Later” campaigns. *Regulatory:* NIS CG PQC roadmap [18] with 2030/2035 milestones; ANSSI requirement for PQC capability in all Qualification submissions from 2027; BSI TR-02102-1 v2026-01 approving ML-KEM, ML-DSA, SLH-DSA, HQC, FrodoKEM, and Classic McEliece.

High-Impact Activities. (Phase 1) Map cryptographic inventories per NIS CG roadmap requirements; optimise FIPS 203/204/205 algorithms for 8/16-bit microcontrollers; deploy hybrid ML-KEM + ECDH-P384 in cloud-side implementations per ETSI TS 103 744 v1.2.1 [33]. (Phase 2) Mandate hybrid PQC in all new Industrial IoT and financial sector deployments requiring DORA compliance; validate lightweight PQC for sub-100 μ J zero-energy nodes under 6G ISAC testbeds. (Phase 3) Establish PQC-native silicon as the EU procurement default.

Catalysts and Enablers. NIST FIPS 203/204/205 [20][21][22]; ETSI TS 103 744 v1.2.1 hybrid key exchange profiles [33]; ENISA PQC migration guidelines; ECC Strategic Agenda [15]; the IETF LAMPS working group X.509 certificate drafts (in RFC Editor Queue); open-source crypto-agility frameworks (e.g., liboqs, Bouncy Castle PQC).

Disruptors and Systemic Risks. Early CRQC arrival before migration is complete. Performance overhead of PQC on battery-powered IoT devices (ML-KEM demonstrates $\sim 1.5\times$ computational overhead versus ECDH at equivalent security levels on ARM Cortex-M4 [32]). Insufficient cryptographic inventory of legacy OT/IoT systems preventing migration planning. Interoperability gaps between hybrid implementations.

Failsafe Mechanisms. Deploy hybrid cryptographic schemes (classical + PQC) during the transition phase to maintain backward compatibility while providing harvest resistance. Implement crypto-agility frameworks enabling algorithm switching within sub-second latency if vulnerabilities are discovered. Diversify across PQC algorithm families (lattice-based, code-based, hash-based) to avoid single-point cryptanalytic failures.

Transversal Meta-Layer. *Sovereignty:* protects long-term European governmental and industrial secrets from future quantum decryption — the 18 Member State joint statement “Securing Tomorrow, Today” (November 2024) formalises this as a collective sovereignty concern. *Trust:* alignment with GDPR Article 32 requirement for state-of-the-art security

measures. *Sustainability:* lightweight PQC implementations reduce the energy overhead and extend device lifecycles, avoiding premature hardware replacement.

C. P3: Lightweight Cryptography and Physical-Layer Security for the Extreme Edge

Baseline and Target. Extreme edge environments — massive machine-type communication nodes, body-area networks, zero-energy backscatter radios — remain constrained by limited compute, energy, memory, and connectivity, making conventional cryptographic protections impractical. The target is a class of lightweight, adaptive security mechanisms combining efficient cryptography with physical-layer primitives tailored for highly constrained environments, enabling quantum-resistant authentication for devices operating at sub-milliwatt power budgets.

Primary Drivers. *Threat:* the proliferation of mMTC requiring 100+ devices per m^3 with severe energy constraints targeting 40-year battery lifespans or energy-harvesting operation. *Regulatory:* CRA applicability to products with digital elements, including resource-constrained IoT devices.

High-Impact Activities. (Phase 1) Test group-based authentication and early lightweight PQC algorithms on resource-constrained IoT; validate RF fingerprinting beyond controlled lab environments. (Phase 2) Hardware-software co-design for neuromorphic/PIM AI-assisted security on EU open-hardware platforms. (Phase 3) Deploy zero-energy physical-layer authentication as a standard identity mechanism for battery-less mMTC nodes.

Catalysts and Enablers. NIST lightweight cryptography efforts; 6G and O-RAN research programmes under SNS JU [23]; ETSI SmartBAN standardisation for body-area networks; European experimentation platforms for edge and wireless security.

Disruptors and Systemic Risks. Variable reliability of PHY-based techniques under environmental changes; device heterogeneity; danger of deploying under-validated mechanisms in safety-critical systems; immaturity of deployment standards.

Failsafe Mechanisms. Layered combinations of lightweight cryptography and higher-layer verification; adaptive confidence thresholds; dynamic fallback to localised trust boundaries using purely physical-layer signatures when upper-layer cryptographic processing exceeds the available energy budget.

Transversal Meta-Layer. *Sovereignty:* secures the far edge of European digital infrastructure where conventional approaches cannot reach. *Sustainability:* energy-aware security is a precondition for scalable extreme-edge deployment without disproportionate environmental cost.

D. P4: Security Interoperability via Simpl Middleware

Baseline and Target. European cloud-edge ecosystems remain fragmented across proprietary vendor silos with incompatible security policies, authentication mechanisms, and data exchange protocols. The Simpl programme — funded through a €41M development contract to the Sovereign-X consortium (led by Eviden Belgium, awarded February 2024) and a €65M

second tender to InfrateX (led by Sopra Steria + NTT DATA) — has produced an iterative MVP available since January 2025 with enhanced releases in June and December 2025, hosted on EU GitLab (code.europa.eu/simpl/simpl-open) [16]. The target is a unified European “Trust Fabric” enabling secure, federated interoperability across heterogeneous providers through standardised security Minimal Interoperability Mechanisms (MIMs) within the Simpl stack. However, OASC’s MIM6 (Security management) remains at “Work item” status with no formally published specification as of April 2026.

Primary Drivers. *Regulatory:* Data Act (Regulation (EU) 2023/2854, applicable from 12 September 2025) [12] mandating interoperability and portability; NIS2 requirements for coordinated incident response across federated infrastructure. *Market:* breaking vendor lock-in — the primary barrier to EU provider competitiveness is proprietary API ecosystems, not technical superiority.

High-Impact Activities. (Phase 1) Launch Simpl security MIM pilots in at least three Common European Data Spaces (Manufacturing, Health, Energy). (Phase 2) Deploy automated policy translation via standardised Simpl MIMs; achieve adoption as ratified CEN/CENELEC standards referenced in the EU Cloud Rulebook. (Phase 3) Establish Simpl-based security as the universal standard for Data Space operation.

Catalysts and Enablers. Digital Europe Programme funding for Simpl development; Gaia-X Trust Framework 3.0 “Danube” (November 2025); CEN-CLC/JTC 25 (established September 2024, chaired by Sebastian Steinbuss/IDSA) for data-space standards; Eclipse Dataspace Protocol nearing v1.0 release; IPCEI-CIS ICRA Reference Architecture v2.0 (January 2026).

Disruptors and Systemic Risks. Vendor resistance to open standards; weakest-link security failures in federated systems where one provider’s compromise cascades; insufficient incentives for interoperability; uneven maturity across sectors and Member States.

Failsafe Mechanisms. Capability to autonomously unfederate or isolate nodes that fail to meet the shared security baseline; Zero-Trust architectures where federation does not imply implicit trust; cryptographic attestation of policy enforcement.

Transversal Meta-Layer. *Sovereignty:* breaking hyper-scaler lock-in (AWS, Azure, GCP hold ~70% of the EU cloud market [10]; European-headquartered providers collectively hold ~15%, stable since 2022 but down from 29% in 2017). *Trust:* transparent, verifiable security mechanisms enabling cross-border trust.

E. P5: Supply Chain Verification and Secured Infrastructure

Baseline and Target. Current digital infrastructures depend on complex, opaque, and globally distributed supply chains. CRA Annex I Part II will mandate static SBOM disclosure from December 2027 [2]; NIS2 Article 21(2)(h) makes supply chain risk assessment mandatory for essential entities [1]; DORA Article 28 requires continuous ICT third-party risk monitoring [11] — operationalised through the first 19 Critical ICT Third-Party Provider designations on 18 November 2025 [34]. The target is verifiable and continuously assessable

supply chains spanning hardware, software, firmware, and service dependencies, progressing from static disclosure (Phase 1) through runtime verification (Phase 2) to real-time ZKP-attested provenance (Phase 3).

Primary Drivers. *Threat:* rising software supply chain attacks (SolarWinds-class); disaggregation of telco clouds introducing extreme multi-vendor complexity. *Regulatory:* simultaneous CRA SBOM, NIS2 supply chain risk assessment, and DORA ICT third-party monitoring obligations.

High-Impact Activities. (Phase 1) Submit CRA-mandated static SBOM declarations; initiate Digital Product Passport pilot programmes; advance ZKP verification toward demonstrator readiness. (Phase 2) Deploy Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure; validate ZKP batch verification operationally. (Phase 3) Operationalise ZKP-verified supply chain attestation at real-time transaction scale.

Catalysts and Enablers. CRA standardisation request M/606 [2]; CEN-CLC/JTC 13/WG 9 horizontal standards (EN 40000 series); EU Ecodesign Regulation Digital Product Passport requirements for ICT hardware; DORA Delegated Regulation (EU) 2024/1502 for CTPP assessment.

Disruptors and Systemic Risks. Global concentration of manufacturing dependencies; limited multi-tier auditability; high complexity of continuously verifying dynamic supply chains; SME exclusion from compliance pathways.

Failsafe Mechanisms. Graceful isolation of unverified components into quarantined network slices; multi-vendor diversification; trusted fallback components; continuous monitoring for supply chain anomalies.

Transversal Meta-Layer. *Sovereignty:* verifiability tied to accountability and procurement governance. *Trust:* the DORA CTPP designation of 19 providers (including AWS, Google Cloud, Microsoft, SAP, Deutsche Telekom, Orange, SWIFT [34]) demonstrates the operational reach of EU supply chain oversight.

F. P6: Cognitive SecOps (LLM SecOps) and Automated Threat Intelligence

Baseline and Target. Security operations remain heavily manual, fragmented, and overloaded by alert volumes and analyst scarcity. US-platform SIEMs dominate the EU market, with no EU-headquartered vendor in the Gartner Magic Quadrant Leaders quadrant (October 2025) [17]. Foundation models for SOC copilot functions are US-origin. The most concrete European sovereign security AI initiative is Thales’ GenAI4AID, built on Mistral AI models; the OpenEuroLLM consortium (launched February 2025, €37.4M budget) has no dedicated security model stream [35]. The target is a cognitive SecOps model where EU-sovereign language models and AI-driven reasoning systems augment detection, triage, investigation, and response while remaining controllable, auditable, and secure under AI Act Article 9 conformity assessment.

Primary Drivers. *Threat:* weaponisation of generative AI for polymorphic malware, automated exploit discovery, and social engineering at scale. *Regulatory:* AI Act Articles 9, 13, and 15 creating mandatory requirements for autonomous

security AI classified as high-risk systems [3]; a 24–36 month regulatory gap where obligations precede conformity assessment capacity.

High-Impact Activities. (Phase 1) Integrate commercial LLMs into SOC analyst workflows for alert triage and MITRE ATT&CK technique mapping; initiate training of at least one EU-sovereign SLM ($\leq 10B$ parameters) on GDPR-compliant security corpora using EuroHPC compute. (Phase 2) Deploy EU-sovereign SLMs at the edge for real-time SOC analytics without cloud inference dependency; publish a CEN/CENELEC-standardised adversarial robustness test suite. (Phase 3) Deploy swarm-based agentic AI for decentralised threat intelligence.

Catalysts and Enablers. EuroHPC AI Factories — 19 confirmed across 16 Member States with €2.7B cumulative funding; IT4LIA (CINECA), PIAST (Poznań), and MeluXina-AI (Luxembourg) explicitly list cybersecurity as a vertical [36]. ETSI GS ZSM 009 (closed-loop automation) [37]; ECCC Strategic Agenda; Sekoia.io/HarfangLab French Open XDR Platform consortium.

Disruptors and Systemic Risks. Hallucinations and automation bias; adversarial prompt manipulation; data leakage into models; no standardised adversarial robustness test suite for security AI (the single most consequential standards gap in this cluster); overreliance on opaque AI behaviour in critical contexts.

Failsafe Mechanisms. Human-in-the-loop validation for Tier-1 critical actions; bounded autonomy with ETSI GS ZSM 009 governance; “shadow-mode” rollback if autonomous agents exhibit unstable control loops; audit logging and policy-constrained orchestration.

Transversal Meta-Layer. *Sovereignty:* reducing ~80% US-platform dependency in enterprise SIEM. *Trust:* AI Act compliance-by-design, explainability, and accountability for autonomous security decisions.

G. P7: Pan-European NaaS Federation (“Schengen for Cloud-Edge/AI”)

Baseline and Target. European digital infrastructures remain fragmented across providers, jurisdictions, and operational models. CAMARA’s Fall 2025 meta-release comprises 60 APIs (10 stable) [27]; GSMA Open Gateway has reached 86 operator groups representing 300+ networks and 80% of global mobile connections [38]. CEF Digital Call 4 selected 12 5G corridor projects in Q4 2025. However, cross-border edge compute roaming does not exist at operational scale. The target is a trusted pan-European NaaS federation enabling secure, policy-aligned workload migration across operator boundaries.

Primary Drivers. *Threat:* the “Sovereignty Cliff” — lack of a true Digital Single Market prevents European operators from scaling to compete with US/China hyperscalers. *Regulatory:* the Digital Networks Act proposal (COM(2026) 16, adopted January 2026) signals EC intent for network federation, though it does not mandate specific API standards.

High-Impact Activities. (Phase 1) Deploy harmonised CAMARA APIs in 2–3 bilateral 5G corridor agreements. (Phase 2) Scale cross-border roaming to ≥ 5 EU operator pairs;

operationalise “Data Visa” mechanisms for dynamic AI model migration. (Phase 3) Operate the EU NaaS federation as a commercially viable unified jurisdiction for data workloads.

Catalysts and Enablers. IPCEI-CIS (€2.6B total, 19 direct participants, 12 Member States) [39]; CAMARA consortium [27]; SNS JU (€630M cumulative EU funding, 100 projects) [23]; Aduna (Ericsson/operator JV) as primary aggregator.

Disruptors and Systemic Risks. Inconsistent governance across Member States; operator misalignment; hyperscaler platform dominance; US CLOUD Act jurisdictional conflict.

Failsafe Mechanisms. Graceful degradation protocols redirecting workloads to national infrastructure; regional autonomy zones; multi-operator fallback routing.

Transversal Meta-Layer. *Sovereignty:* the “Schengen for Cloud” legal framework would resolve structural CLOUD Act conflict. *Trust:* legal interoperability and equitable distribution of strategic infrastructure value.

H. P8: Confidential Computing and Privacy-Preserving AI Pipelines

Baseline and Target. Sensitive data processing across cloud-edge-AI pipelines remains exposed to platform-level trust assumptions. Intel TDX, AMD SEV-SNP, and ARM TrustZone dominate all operational TEE deployments; EU-origin alternatives (RISC-V Keystone [29], OpenTitan [28]) remain at research/prototype stage. The target is end-to-end privacy-preserving AI pipelines using TEEs, homomorphic encryption, and verifiable processing guarantees, progressively migrating to EU-origin hardware as Cluster 1 matures.

Primary Drivers. *Regulatory:* GDPR Article 25 (data protection by design) [14]; AI Act trustworthiness requirements; EUCS (when adopted) certification for confidential cloud services. *Market:* protection of industrial IP during federated training across multi-provider Data Spaces.

High-Impact Activities. (Phase 1) Design pilot architectures with TEE abstraction layers; initiate FHE benchmarking for edge-viable use cases. (Phase 2) Deploy RISC-V Keystone TEEs in multi-provider Data Spaces alongside Intel TDX infrastructure. (Phase 3) Operationalise FHE end-to-end from IoT sensor to HPC compute node.

Catalysts and Enablers. European Data Space programmes; EUCS certification development; Chips JU RISC-V TEE development; GDPR-aligned data governance frameworks.

Disruptors and Systemic Risks. Hardware dependence on non-European TEE vendors throughout Phase 1; performance overhead of FHE; side-channel vulnerabilities in current TEE implementations; limited cross-vendor TEE attestation interoperability.

Failsafe Mechanisms. Hybrid privacy architectures; selective workload confinement; limit processing to local federated learning nodes if edge TEE integrity cannot be verified.

Transversal Meta-Layer. *Sovereignty:* Intel/AMD/ARM TEE dependency is ~100% for operational deployments — this is a critical sovereignty gap. *Trust:* verifiable data protection strengthens AI adoption legitimacy.

I. P9: Resilient-by-Design Infrastructure and Autonomous “Island Mode”

Baseline and Target. Many critical digital infrastructures assume persistent connectivity, stable central coordination, and continuous service dependencies. The target is resilient-by-design infrastructure capable of autonomous local operation, graceful degradation, and secure continuation under disruption through “Island Mode” — sustaining a minimum of 72 hours of autonomous operation under network partition or orchestration-plane compromise.

Primary Drivers. *Threat:* escalating cyberattacks, extreme weather, and geopolitical tensions causing cascading failures. *Regulatory:* NIS2 business continuity mandates for essential entities [1]; DORA Article 11 business continuity requirements for financial entities [11]; SNS JU SRIA requirements for 6G resilience and AI-RAN [23].

High-Impact Activities. (Phase 1) Validate Island Mode soft-state protocols in 5G testbeds aligned with SNS JU objectives; develop autonomous fallback mechanisms for edge nodes lacking survivability modes. (Phase 2) Achieve standardised certification of 72-hour Island Mode under 3GPP/ETSI fallback API profiles in EU 5G cross-border corridor pilots. (Phase 3) Mandate validated 72-hour autonomous Island Mode across all NIS2-essential and DORA-critical edge deployments.

Catalysts and Enablers. DORA [11]; SNS JU 6G resilience projects (HORSE, ROBUST-6G, ELASTIC) [23]; civil protection and critical infrastructure programmes; resilience-oriented public procurement.

Disruptors and Systemic Risks. Centralised architectural legacies; poor failover design; excessive cloud dependence; insufficient integration between safety, security, and operational resilience engineering.

Failsafe Mechanisms. This capability inherently depends on redundant trust anchors, segmented autonomy zones, offline continuity functions, and tamper-resistant local decision logic. Zero-energy backup layers or localised LPWAN micro-slices maintain a “minimal viable network” for life-saving telemetry if standard edge power and computing fail.

Transversal Meta-Layer. *Sovereignty:* sovereign infrastructure must be dependable under crisis. *Resilience:* P9 directly operationalises the “cyber-resilience” concept at the infrastructure level.

J. P10: Proactive Cyber Threat Management and Defence against APTs

Baseline and Target. Threat management remains largely reactive, with limited anticipation of coordinated, stealthy, and persistent adversaries. Average APT dwell times remain at 150–200 days in many European organisations. The target is a proactive defence posture combining predictive threat intelligence, attack path analysis, adversary emulation mapped to the MITRE ATT&CK framework, and adaptive Moving Target Defence, achieving dwell times below 72 hours in Phase 2 and below 24 hours in Phase 3 across all NIS2-essential deployments.

Primary Drivers. *Threat:* nation-state actors blending cyber espionage with organised crime; exploitation of complex supply chains (SolarWinds-class attacks); stealthy techniques including memory-only execution and abuse of legitimate administrative tools. *Regulatory:* NIS2 Article 21(2)(b) incident handling requirements [1].

High-Impact Activities. (Phase 1) Apply ML-based User and Entity Behaviour Analytics for APT IoC detection; target dwell time reduction below 72 hours in at least one certified NIS2-essential entity. (Phase 2) Deploy deception infrastructure alongside AI detection for model calibration; orchestrate autonomous triage under ETSI GS ZSM 009 governance [37]. (Phase 3) Operationalise Moving Target Defence continuously shuffling IP addressing, network topology, and software diversity.

Catalysts and Enablers. ENISA threat intelligence efforts; CSIRT and SOC cooperation structures; MITRE ATT&CK framework integration into incident management; pan-European CTI sharing.

Disruptors and Systemic Risks. Intelligence fragmentation; poor information-sharing incentives; adversarial adaptation outpacing defensive innovation; organisational reluctance to invest in proactive capabilities before incidents occur.

Failsafe Mechanisms. Autonomous real-time micro-segmentation (Zero-Trust) to quarantine compromised components even if the specific exploit is unknown; layered detection logic; deception and containment zones; rapid recovery playbooks.

Transversal Meta-Layer. *Sovereignty:* European strategic preparedness requires sovereign threat intelligence independent of US-origin feeds. *Governance:* information sharing requires trustworthy automation and proportionate handling of intelligence-driven interventions.

VII. PRIORITY CLUSTERS AND TRANSITION MATRIX

The ten capabilities are consolidated into three executive priority clusters analysed across two dimensions per phase: (i) Technical Actions and Deployment Targets and (ii) Governance, Policy and Standards Milestones. Each phase carries a distinct mandate reflecting the characteristic transition from regulatory enforcement (Phase 1) through industrial scale-up (Phase 2) to sovereign leadership (Phase 3).

A. Priority Cluster 1: Sovereign Trusted Infrastructure

This cluster establishes the cryptographically verifiable trust foundation for the European cloud-edge-IoT continuum. It unifies five capabilities: P1 (Hardware Roots of Trust), P2 (PQC Agility), P3 (Lightweight Cryptography and PhySec), P8 (Confidential Computing), and P9 (Resilient-by-Design Infrastructure). This cluster provides the hardware and cryptographic assurances that Clusters 2 and 3 progressively consume as their own capabilities mature.

The cluster addresses a single architectural premise: federated security policies (Cluster 3) and autonomous defence capabilities (Cluster 2) gain operational credibility in proportion to the maturity of the execution substrate on which they run. That substrate must guarantee three properties: hardware

integrity verifiable through open-source silicon roots of trust; quantum-resistant confidentiality through crypto-agile schemes aligned with FIPS 203/204/205 [20][21][22]; and autonomous resilience enabling continued operation under network or system-level compromise.

The shared critical dependency across this cluster is European sovereign fabrication capacity under the EU Chips Act [8]. Hardware Roots of Trust, PQC Agility, Lightweight Cryptography, and Confidential Computing all require RISC-V-based secure elements manufactured on EU-controlled process nodes to reach their Phase 2 targets. This fabrication bottleneck — underscored by the ECA's contested 20% target [9] and Intel Magdeburg's cancellation — is the single most consequential cross-priority risk in the roadmap.

1) Phase 1: Foundations and Pilot Actions (2026–2030): Sovereignty Level: High Foreign Dependency

Technical Actions. Fund RISC-V secure element projects targeting EUCC 'High' under Horizon Europe Cluster 3; target Fragmented Piloting by 2030. Validate Island Mode soft-state protocols in 5G testbeds aligned with SNS JU SRIA objectives [23]. Optimise FIPS 203/204/205 algorithms for 8/16-bit microcontrollers. Submit Keystone [29] and Open-Titan [28] prototypes for EUCC pre-evaluation. Validate RF fingerprinting in representative field conditions. Initiate FHE benchmarking for edge-viable use cases, recognising that operational deployment is not achievable before Phase 2. Establish 3–5 Horizon Europe consortia operating RISC-V secure element pilots in controlled critical infrastructure demonstrators. Design all pilot architectures with explicit TEE abstraction layers to avoid lock-in to Intel TDX, AMD SEV-SNP, or ARM TrustZone, which will dominate all operational TEE deployments throughout this phase. Initiate PQC deployment in cloud-side software implementations; far-edge IoT will run classical cryptography exclusively throughout Phase 1. Develop autonomous fallback mechanisms for edge nodes lacking survivability modes meeting DORA [11] or NIS2 [1] business continuity thresholds.

Governance and Standards. Align all R&I outputs with the regulatory enforcement timeline: NIS2 Article 21(2)(e) cryptographic policies (legal obligation since October 2024, though transposition remains incomplete in approximately one-third of Member States [4]); CRA Articles 10–11 hardware security certification (conformity assessment body notification from June 2026, full application December 2027 [2]); AI Act Articles 9 and 15 hardware integrity requirements (applicability from August 2026 under the default timetable, potentially deferred to December 2027 under the Digital Omnibus [5]). Adopt the NIS CG PQC roadmap [18] as the baseline for all new cryptographic deployments. Engage in CEN/CENELEC JTC 13 hardware security standards drafting. Prioritise achieving EUCC 'High' certification for an EU-origin RISC-V secure element before end of phase — this is the critical regulatory gate. Note that EUCC 'High' has already been achieved by Infineon, NXP, and STMicroelectronics for proprietary secure elements [6]; the Phase 1 target is certification of an *open-source, EU-auditable* design. Initiate EU Chips Act fabrication investment targeting pre-production milestones. Establish strategic stockpiling and second-source

agreements as interim mitigation against foundry access disruption.

2) Phase 2: Scale-Up, Industrial R&D and Standardisation (2030–2035): Sovereignty Level: Emerging EU Ecosystem

Technical Actions. Bring the first EUCC 'High' certified EU-manufactured RISC-V secure elements into production. Deploy in energy substations and telecom critical infrastructure in at least five Member States. Target Standardised Integration for Hardware RoT and Island Mode by 2035. Validate hybrid ML-KEM + ECDH-P384 key encapsulation at Industrial IoT scale. Demonstrate lightweight PQC for sub-100 μ J zero-energy nodes under 6G ISAC testbeds. Deploy RISC-V Keystone TEEs operationally in multi-provider Data Spaces alongside Intel TDX infrastructure, validating cross-vendor TEE attestation interoperability. Achieve standardised certification of 72-hour Island Mode under 3GPP/ETSI fallback API profiles in at least three EU 5G cross-border corridor pilots under SNS JU governance. Mandate hybrid PQC schemes as standard in all new Industrial IoT and financial sector deployments requiring DORA compliance.

Governance and Standards. Secure the definitive Phase 2 policy gate: EUCC 'High' certification of the first EU-manufactured RISC-V secure element, transforming applied research into a market-accessible, legally certifiable product. Adopt NIST FIPS 203/204/205 into ETSI TS profiles for EU deployment contexts. Press ENISA to publish a mandatory PQC migration timeline referenced in NIS2 sector-specific implementing acts. Drive CEN/CENELEC hardware security standards through ratification. Target 15–30% market share for EU-designed secure elements in new critical infrastructure procurement. Establish second-source fabrication agreements between EU foundries.

3) Phase 3: Strategic Autonomy and Sovereign Leadership (2035–2040): Sovereignty Level: Full Sovereign Autonomy

All five priorities target Ubiquitous Operation. Establish EU-designed RISC-V secure elements as the hardware default across all new EU ICT products, with 60% or more of critical infrastructure edge nodes using EUCC 'High' certified EU-designed elements. Operationalise FHE end-to-end from IoT sensor to HPC compute node. Deploy zero-energy physical-layer authentication via RF fingerprinting as a standard identity mechanism for battery-less mMTC nodes. Mandate validated 72-hour autonomous Island Mode survivability across all NIS2-essential and DORA-critical edge deployments. Make EUCC 'High' certification mandatory for all new EU ICT products without waiver. Establish PQC-native silicon as the EU procurement default. Achieve adoption of at least one EU hardware security standard in ISO/IEC or ITU-T international standardisation. R&I investment orients toward neuromorphic secure compute, post-second-generation quantum hardware, and post-CMOS security architectures.

B. Priority Cluster 2: Cognitive Security and Autonomous Cyber Defence

This cluster operationalises artificial intelligence as both the primary instrument and a critical attack surface within the European cyber defence architecture. It encompasses P6 (LLM-SecOps) and P10 (Proactive APT Management), together with

cross-layer dimensions governing AI system integrity: Model Integrity and DataBOMs (CL1.2), Dynamic Supply Chain Verification (CL1.3), and AI Trustworthiness and Algorithmic Accountability (CL4.7).

The cluster is structured around three inseparable sub-dimensions. First, AI as a security instrument: deploying sovereign European foundation models, SLMs, and edge-native agents for autonomous SOC operations and real-time Moving Target Defence against APT campaigns mapped to the MITRE ATT&CK framework. Second, AI as an adversary amplifier: recognising that generative capabilities are simultaneously available for polymorphic malware generation and automated exploit discovery. Third, AI as target: securing training pipelines and inference endpoints against data poisoning, model evasion, and prompt injection, in direct compliance with AI Act Article 9 requirements for high-risk system robustness [3].

As Cluster 1 matures its TEE and hardware attestation capabilities through each phase, Cluster 2 progressively deploys autonomous agents onto increasingly trustworthy substrates — from cloud-hosted copilots (Phase 1) to edge-native autonomous guardians (Phase 3). Delays in any one cluster constrain the operational credibility of the other two at each subsequent phase gate.

1) Phase 1: Foundations and Pilot Actions (2026–2030): Sovereignty Level: High Reliance on US Foundation Models

Technical Actions. Integrate commercial LLMs into SOC analyst workflows for alert triage and MITRE ATT&CK technique mapping; target Siloed AI Copilots by 2030. Route NWDAF telemetry pipelines into centralised AI analytics in 3–5 5G SA pilot deployments [26]. Apply ML-based User and Entity Behaviour Analytics for APT IoC detection. Target APT dwell time reduction below 72 hours in at least one certified pilot deployment in a named NIS2-essential entity. Develop adversarial robustness testing methodologies for prompt injection, model evasion, and DataBOM verification; no operational certification framework exists at phase start. Acknowledge that no certified autonomous containment agent will reach operational deployment during Phase 1.

Governance and Standards. Prepare for AI Act Articles 9, 13, and 15 (applicability from August 2026): mandatory requirements for autonomous security AI classified as high-risk. Address the 24–36 month regulatory gap where obligations precede conformity assessment body operational readiness [5]. Adopt ETSI GS ZSM 009 [37] as the governance reference for all autonomous SOC pilots. Operationalise NIS2 Article 21(2)(b) incident handling across pilot SOC federations. Prioritise development of a standardised adversarial robustness test suite under CEN/CENELEC or ETSI TC CYBER — the single most consequential standards gap in this cluster. Initiate training of at least one EU-sovereign SLM ($\leq 10B$ parameters) on GDPR-compliant security corpora using EuroHPC compute [36].

2) Phase 2: Scale-Up, Industrial R&D and Standardisation (2030–2035): Sovereignty Level: Sovereign EU Edge LLMs Emerging

Technical Actions. Deploy EU-sovereign SLMs trained on MITRE ATT&CK-aligned corpora at the edge for real-time

SOC analytics without cloud inference dependency. Enable Guardian nodes to execute autonomous Zero-Trust reconfigurations per NIST SP 800-207 [19]. Orchestrate autonomous incident triage under ETSI GS ZSM 009 governance. Target APT dwell time ≤ 72 hours in certified critical infrastructure deployments. Deploy deception infrastructure (honeynets, credential canaries) alongside AI detection for model calibration ground truth.

Governance and Standards. Secure the definitive Phase 2 policy gate: operationalisation of a standardised AI Act conformity assessment framework for autonomous security agents. Publish a CEN/CENELEC-standardised adversarial robustness test suite. Formally reference MITRE ATT&CK in NIS2 sector-specific incident handling guidance. Activate EuroHPC AI Factory compute contracts for EU-sovereign security model training [36]. Target EU-sovereign SLMs covering $\geq 30\%$ of critical infrastructure SOC deployments in regulated sectors.

3) Phase 3: Strategic Autonomy and Sovereign Leadership (2035–2040): Sovereignty Level: Global Leadership in Trustworthy AI Security

All sub-dimensions target Ubiquitous Operation. Deploy swarm-based agentic AI defending the EU Cloud-Edge Continuum via self-evolving, decentralised threat intelligence. Operationalise Moving Target Defence continuously shuffling IP addressing, network topology, and software diversity; target APT reconnaissance success rates below 5% in certified environments and dwell times ≤ 24 hours across all NIS2-essential deployments. Make adversarial robustness validation continuous and fully automated. EU-sovereign AI security tools target $\geq 50\%$ of new critical infrastructure SOC deployments. Achieve international adoption of EU AI Act frameworks for trustworthy autonomous security agents in at least one non-EU partner jurisdiction. R&I orients toward quantum-native AI security architectures, neuromorphic threat detection, and AI-to-AI adversarial dynamics.

C. Priority Cluster 3: Federated Operational Sovereignty

This cluster creates the governance and interoperability fabric that transforms individually sovereign components into a collectively sovereign European computing continuum. It encompasses P4 (Security Interoperability via Simpl), P5 (Supply Chain Verification), and P7 (Pan-European NaaS Federation).

The cluster addresses three complementary mechanisms. First, federated security interoperability: establishing standardised MIMs within the Simpl stack [16] for provider-agnostic policy enforcement across Common European Data Spaces, in compliance with Data Act portability mandates [12]. Second, supply chain integrity: deploying Runtime SBOMs, Digital Product Passports, and cryptographic proof-of-execution attestation, as required under CRA Annex I [2]. Third, pan-European compute federation: harmonising Network APIs through CAMARA [27] and establishing cross-border edge roaming interfaces.

Supply Chain Verification bridges Clusters 1 and 3: its hardware attestation dimension depends on the RISC-V secure elements that Cluster 1 matures, while its runtime verification dimensions constitute governance prerequisites for the

federated trust fabric. Without a “Schengen for Cloud” legal framework and enforceable cross-border compliance APIs, compute federation remains commercially fragile regardless of technical standardisation.

1) Phase 1: Foundations and Pilot Actions (2026–2030): Sovereignty Level: Mixed — Emerging EU Ecosystem / Hyperscaler Dominance

Technical Actions. Launch Simpl security MIM pilots in at least three Common European Data Spaces (Manufacturing, Health, Energy) with initial manual policy coordination; target Fragmented Piloting by 2030. The European Health Data Space entered into force on 26 March 2025 [40] with staggered primary-use applications. Deploy harmonised CAMARA Network APIs in 2–3 bilateral 5G corridor agreements; cross-border edge compute roaming does not exist at operational scale and will not during this phase. Submit CRA-mandated static SBOM declarations as point-in-time compliance artefacts. Initiate Digital Product Passport pilot programmes. Advance ZKP verification for supply chain attestation toward demonstrator readiness; runtime verification is not achievable in this phase.

Governance and Standards. Prepare for simultaneous legislative activation: CRA SBOM obligations (enforcement December 2027) [2]; NIS2 Article 21(2)(h) mandatory supply chain risk assessment [1]; DORA Article 28 continuous ICT third-party monitoring (operationalised through the 19 CTPPs designated November 2025 [34]); Data Act interoperability obligations (applicable from September 2025) [12]. Secure the Simpl-Open iterative MVP as the Phase 1 technical trigger for Security Interoperability — noting that Simpl-Open is an iterative MVP (available since January 2025 with enhanced releases in June and December 2025), not a versioned product release [16]. Engage in IPCEI-CIS deployment phase governance. Address the critical compliance fragmentation risk that disproportionately burdens SMEs. Hyperscalers retain approximately 70% of the EU cloud market through proprietary API ecosystems [10]; the primary barrier to EU provider competitiveness is API lock-in, not technical superiority.

2) Phase 2: Scale-Up, Industrial R&D and Standardisation (2030–2035): Sovereignty Level: Emerging EU Ecosystem

Technical Actions. Deploy automated policy translation via standardised Simpl MIMs across the majority of active Common European Data Spaces; target Standardised Integration. Deploy Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure deployments in at least five Member States. Validate ZKP batch verification operationally. Automatically quarantine unverified components in isolated network slices pending attestation. Scale CAMARA cross-border roaming to ≥ 5 EU operator pairs. Operationalise “Data Visa” mechanisms for dynamic AI model migration without manual legal review.

Governance and Standards. Secure the definitive Phase 2 policy gate: adoption of Simpl security MIMs as ratified CEN/CENELEC standards, referenced as mandatory compliance requirements in the EU Cloud Rulebook. Without ratified MIM standards referenced in procurement legislation, Simpl-based interoperability remains voluntary without enforcement power. Enforce mandatory Digital Product Passports under

the EU Ecodesign Regulation for ICT hardware categories. Complete DORA continuous ICT third-party monitoring operationalisation. Target $\geq 30\%$ non-hyperscaler share in new regulated-sector cloud procurement. Foreign hyperscaler market share declines for the first time from the approximately 70% Phase 1 baseline [10] — noting that France is the only Member State with measurable procurement shift data (DINUM 2025 bilan: 70% European providers market-wide, 99% within the State perimeter [41]).

3) Phase 3: Strategic Autonomy and Sovereign Leadership (2035–2040): Sovereignty Level: Full Federated Sovereign Autonomy

All three priorities target Ubiquitous Operation. Establish Simpl-based security as the universal standard for cross-provider Data Space operation. Operationalise ZKP-verified supply chain attestation at real-time transaction scale. Operate the EU NaaS federation as a commercially viable unified jurisdiction for data workloads. Enact the “Schengen for Cloud” legal framework, resolving US CLOUD Act jurisdiction conflicts. Make the EU Cloud Rulebook mandatory for all new regulated-sector cloud deployments. Non-EU hyperscaler market share in EU regulated cloud procurement falls below 40% for new deployments. Ensure EU supply chain governance standards (Runtime SBOM + ZKP) are referenced in ISO/IEC international standards and adopted by at least two partner jurisdictions.

D. Cross-Cutting Enablers

The following dependencies span two or more clusters and require dedicated governance to avoid becoming systemic bottlenecks.

EU Chips Act Sovereign Fabrication. Clusters 1 and 3 both depend on EU-controlled RISC-V secure element fabrication reaching production capacity by Phase 2. Five official Chips JU pilot lines are now operational (NanoIC, FAMES, APECS, PIXEurope, WBG) with combined EU and national funding of €3.7B [8]. However, no pilot line is dedicated to secure element fabrication, and restriction of access to any major non-EU foundry would immediately impair EU critical infrastructure attestation capability. Mitigation: strategic stockpiling, second-source agreements, and dedicated EU Chips Act pilot lines for security-critical components.

AI Act Conformity Assessment Infrastructure. Cluster 2 faces a 24–36 month regulatory gap where AI Act obligations precede the operational readiness of conformity assessment bodies capable of certifying autonomous security agents. As of March 2026, only 8 of 27 Member States had designated single points of contact for AI Act enforcement [5]. This gap blocks Phase 2 deployment regardless of technical maturity. Mitigation: accelerate CEN/CENELEC adversarial robustness test suite development and activate AI Act sandbox provisions for security-specific use cases.

Runtime SBOM Attestation. Clusters 1 and 3 both require runtime verification of supply chain component integrity. Static SBOM declarations (Phase 1, CRA-mandated) are insufficient for dynamic multi-provider continuum operations. Progression toward cryptographic proof-of-execution and ZKP

batch verification is a shared transition condition for Phase 2–3 maturity.

SME Access and Inclusion. All three clusters identify cost barriers excluding SMEs from sovereign hardware alternatives, AI-powered SOC tooling, and Simpl MIM compliance. Digital Europe Programme co-funding should target shared-infrastructure pathways and subsidised certification programmes accessible to smaller actors.

Standards-Setter Ambition. All three clusters converge on the same Phase 3 strategic objective: transitioning the EU from standards-taker to standards-setter. The evidence threshold is adoption of at least one EU-origin standard in ISO/IEC or ITU-T international standardisation, and formal adoption by at least two non-EU partner jurisdictions. This requires coordinated diplomatic engagement beginning in Phase 1.

VIII. STRATEGIC RECOMMENDATIONS

The 2040 destination is a secure, sovereign, interoperable, and resilient European computing continuum that is trusted by design and resilient by default. Based on the analysis in this paper, six strategic actions are required.

First, Europe should invest in **sovereign secure silicon** by combining Horizon Europe support with the EU Chips Act to advance open-source RISC-V secure elements and pilot fabrication lines capable of reaching EUCC ‘High’ assurance. The EUCC ‘High’ certifications achieved by STMicroelectronics, NXP, and Infineon in 2025 [6] demonstrate that the EU certification framework is operational; the remaining gap is *open-source, auditable* designs on *EU-controlled* fabrication.

Second, Europe should **operationalise PQC migration** through crypto-agility middleware for edge and IoT environments, aligned with the NIS CG PQC roadmap’s 2030/2035 milestones [18] and ETSI TS 103 744 hybrid key exchange profiles [33]. Hybrid cryptographic deployment should become the default for new critical infrastructure.

Third, Europe should **enforce security interoperability through Simpl and security MIMs**. MIM6 (Security management) must progress from its current “Work item” status to a formally published specification, then to ratified CEN/CENELEC standard referenced in the EU Cloud Rulebook, to achieve procurement-level enforcement.

Fourth, Europe should move toward **continuous certification**. The EUCC scheme [6] provides the foundation; the EUCS scheme must be unblocked — likely through the 2026 Cybersecurity Act revision (COM(2026) 13) [7] — to extend continuous assurance to cloud services.

Fifth, the EU ecosystem should establish a **federated SOC and CTI capability** building on the DORA CTPP oversight framework [34] and cross-border CSIRT cooperation to support pan-European situational awareness.

Sixth, Europe should adopt an **SME inclusion mandate** so that compliance with NIS2, the CRA, and the AI Act does not become a structural barrier to innovation. Tiered certification models, shared cyber range access, and reusable compliance tooling are necessary to avoid a two-speed European cybersecurity ecosystem.

IX. DISCUSSION: VALIDATION, LIMITATIONS, AND METHODOLOGICAL POSITIONING

A. Validation

The roadmap is not treated as a closed analytical product but as an evolving strategic artefact that benefits from expert review, stakeholder consultation, and iterative revision. The validation process draws on expert workshops conducted within the NexusForum and EuCloudEdgeIoT Working Group on Cybersecurity, stakeholder consultation with national cybersecurity agencies and industry representatives, working group review cycles, and policy dialogue with European Commission services. Validation addresses five areas: completeness and relevance of the architectural decomposition, clarity and non-overlap of capability formulations, coherence of priority clustering, credibility of transition conditions, and reasonableness of Sovereignty Level assessments.

B. Limitations

Several limitations should be acknowledged. The architectural decomposition reflects a particular analytical perspective and may not capture all relevant dimensions — particularly those emerging from the intersection of cybersecurity with adjacent domains such as energy policy, industrial policy, or international trade. The seven-dimension profiles rely on expert judgement for assessments of baseline conditions, risk severity, and target feasibility; they do not constitute empirical measurement. The transition conditions sacrifice precision for robustness — they are better suited to strategic orientation than to operational planning. The Sovereignty Level concept resists precise quantification: it is employed as an interpretive strategic lens rather than a metric with a defined scale.

The paper also reflects the ambition–implementation gap that characterises the European regulatory landscape as of 2026. NIS2 transposition failure in Germany, France, Spain, the Netherlands, and Poland — five of the six largest Member States — demonstrates that the implementation gap, not the rule-making pace, is the critical variable for cyber-resilience outcomes [4]. The 20% semiconductor market share target is contested by the European Court of Auditors [9]. EUCS remains unadopted after four years [7]. No harmonised CRA standards have been cited in the Official Journal. These realities constrain the specificity with which Phase 1 governance milestones can be defined and suggest that the roadmap’s greatest value lies not in predicting outcomes but in structuring the coordination challenge.

C. Methodological Positioning

These limitations are consistent with the roadmap’s intended use. The methodology is suitable for strategic agenda setting, expert discussion, cross-priority comparison, and iterative roadmap evolution — even when dependencies, timings, or ecosystem shifts remain uncertain. Its value lies in providing a shared analytical structure within which diverse stakeholders can coordinate research, policy, and investment decisions toward a common set of strategically defined objectives.

X. CONCLUSIONS

The Cloud-Edge-IoT continuum requires Europe to move beyond reactive and fragmented cybersecurity models toward autonomous, federated, and certified Security-by-Design. This paper provided a structured methodology to identify the main security dimensions, derive strategic capabilities, and consolidate them into priorities aligned with resilience and digital sovereignty objectives. The ten capabilities identified — spanning trusted hardware, cryptographic resilience, edge-native security, interoperability, supply chain assurance, AI-enabled operations, federated infrastructure, privacy-preserving computation, resilient continuity, and proactive cyber defence — form an interdependent strategic portfolio that must be developed in parallel.

The 2025–2026 period has produced genuine policy milestones: EUCC ‘High’ certifications for all three major EU secure element manufacturers, a binding PQC migration framework with 2030/2035 horizons, DORA operationalisation through CTPP designation, and emerging Simpl and Data Space infrastructure. Simultaneously, it has revealed persistent gaps: NIS2 transposition delays, EUCS deadlock, contested semiconductor targets, and continued hyperscaler dominance at approximately 70% of the EU cloud market. The ambition–implementation gap identified in this paper is not a reason for pessimism but a structuring observation: it defines where the coordination challenge is most acute and where strategic investment will yield the greatest returns.

Future work should extend this analysis in three directions: empirical validation of Sovereignty Level assessments through quantitative indicators, sector-specific instantiation of the capability profiles for domains such as energy, healthcare, and financial services, and integration of the transition matrix with the evolving EU funding landscape under the next Framework Programme. In this sense, cybersecurity becomes not only a defensive requirement, but a strategic enabler of trustworthy and sovereign European digital infrastructure.

REFERENCES

- [1] European Parliament and Council, “Directive (EU) 2022/2555 on measures for a high common level of cybersecurity across the Union (NIS2 Directive),” *Official Journal of the European Union*, Dec. 2022.
- [2] European Parliament and Council, “Regulation (EU) 2024/2847 on horizontal cybersecurity requirements for products with digital elements (Cyber Resilience Act),” *Official Journal of the European Union*, Nov. 2024.
- [3] European Parliament and Council, “Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI Act),” *Official Journal of the European Union*, Jun. 2024.
- [4] European Commission, “NIS2 Directive transposition in EU countries,” *Shaping Europe’s Digital Future*, 2025. [Reasoned opinions issued to 19 Member States on 7 May 2025.]
- [5] European Parliament Think Tank, “Enforcement of the AI Act,” Mar. 2026. [As of March 2026, 8 of 27 Member States had designated single points of contact. Digital Omnibus on AI (COM(2025) 836) proposes deferral of Annex III high-risk application to December 2027.]
- [6] ENISA, “EUCC — European Cybersecurity Certification Scheme,” Commission Implementing Regulation (EU) 2024/482, applied from 27 February 2025. [First EUCC ‘High’ certificates issued by ANSSI on 1 April 2025 to STMicroelectronics ST54J/K; NXP SN300/JCOP certified July–August 2025; Infineon IFX_CCI certified December 2025 (EUCC-3087-2025-12-0001).]
- [7] European Commission, “Cloud Sovereignty Framework,” published October 2025, updated 17 April 2026. [EUCS candidate scheme remains unadopted as of April 2026. The revised Cybersecurity Act (COM(2026) 13, proposed 20 January 2026) is the likely resolution vehicle.]
- [8] European Parliament and Council, “Regulation (EU) 2023/1781 establishing a framework of measures for strengthening Europe’s semiconductor ecosystem (European Chips Act),” *Official Journal of the European Union*, Sep. 2023. [Five Chips JU pilot lines operational: NanoIC (imec, inaugurated February 2026, €2.5B), FAMES (CEA-Leti, January 2026), APECS (Fraunhofer), PIXEurope, WBG. 30 Chips Competence Centres across 27 Member States plus Norway.]
- [9] European Court of Auditors, “Special Report 12/2025: The European Chips Act,” Apr. 2025. [EU semiconductor value-chain share 9.8% in 2022, projected ~11.7% by 2030. Intel Magdeburg cancelled 24 July 2025.]
- [10] Synergy Research Group, “European cloud infrastructure market share data,” Jul. 2025. [AWS + Azure + GCP hold ~70% of EU market; EU-headquartered providers collectively hold ~15%, down from 29% in 2017.]
- [11] European Parliament and Council, “Regulation (EU) 2022/2554 on digital operational resilience for the financial sector (DORA),” *Official Journal of the European Union*, Dec. 2022.
- [12] European Parliament and Council, “Regulation (EU) 2023/2854 on harmonised rules on fair access to and use of data (Data Act),” *Official Journal of the European Union*, Dec. 2023. [Applicable from 12 September 2025.]
- [13] Microsoft France, testimony before the French Senate, 18 June 2025. [Confirmation that EU data access under US CLOUD Act cannot be guaranteed excluded.]
- [14] European Parliament and Council, “Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data (GDPR),” *Official Journal of the European Union*, Apr. 2016.
- [15] European Cybersecurity Competence Centre (ECCC), “ECCC Strategic Agenda,” 2024.
- [16] European Commission, “Simpl: Cloud-to-edge federations empowering EU data spaces,” *Shaping Europe’s Digital Future*, 2025. [First development contract (€41M) awarded February 2024 to Sovereign-X consortium (Eviden Belgium lead); second tender (€65M) to InfrateX (Sopra Steria + NTT DATA). Simpl-Open MVP available since January 2025 on EU GitLab (code.europa.eu/simpl/simpl-open).]
- [17] Gartner, “Magic Quadrant for Security Information and Event Management,” Oct. 2025. [No EU-headquartered vendor in Leaders quadrant. Palo Alto Networks acquired IBM

QRadar SaaS August 2024; End-of-Life declared 14 April 2025.]

[18] NIS Cooperation Group, “A Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography,” v1.1, 23 Jun. 2025. [Milestones: national roadmaps by December 2026; high-risk critical infrastructure transitioned by December 2030; full completion by December 2035. Builds on Commission Recommendation C(2024) 2393.]

[19] S. Rose et al., “Zero Trust Architecture,” NIST Special Publication 800-207, Aug. 2020.

[20] NIST, “Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM),” FIPS 203, Aug. 2024.

[21] NIST, “Module-Lattice-Based Digital Signature Standard (ML-DSA),” FIPS 204, Aug. 2024.

[22] NIST, “Stateless Hash-Based Digital Signature Standard (SLH-DSA),” FIPS 205, Aug. 2024.

[23] Smart Networks and Services Joint Undertaking, “Strategic Research and Innovation Agenda (SRIA),” SNS JU, 2023. [As of March 2026: €630M cumulative EU funding, 100 projects, €116M allocated to 20 new 6G projects at MWC 2026.]

[24] 3GPP, “Study on security aspects of 3GPP support for Non-Terrestrial Networks (NTN),” 3GPP TR 33.836, Release 17.

[25] G. Dreo et al., “Deliverable D4.4: Cybersecurity Roadmap for Europe,” CONCORDIA Horizon 2020, Dec. 2022.

[26] 3GPP, “Architecture enhancements for 5G System to support network data analytics services,” 3GPP TS 23.288, Release 18.

[27] CAMARA, “Fall 2025 meta-release,” Oct. 2025. [60 APIs: 10 stable, 23 new initial, 27 updates. Security profile built on OpenID Connect Authorization Code Flow and OIDC CIBA.]

[28] lowRISC, “OpenTitan: Open source silicon root of trust,” <https://opentitan.org>, 2024. [Commercial availability via zeroRISC/Nuvoton/Winbond since February 2024; “OpenTitan Integrated” (Rivos) presented at RISC-V Summit Europe, May 2025.]

[29] A. Lee et al., “Keystone: An Open Framework for Architecting TEEs,” in *Proc. 15th European Conf. on Computer Systems (EuroSys)*, Apr. 2020, pp. 1–16, doi: 10.1145/3342195.3387532.

[30] EuroHPC JU, “DARE SGA1,” Grant Agreement 101202459, launched 6 March 2025. [€240M total: €120M JU + €120M Member States. Three chiplets: Openchip Vector Accelerator, Axelera AI AIPU, Codaip General-Purpose Processor.]

[31] NIST, “NIST Selects HQC as Fifth Algorithm for Post-Quantum Encryption,” Mar. 2025. [Code-based backup KEM; draft FIPS anticipated mid-2026.]

[32] NIST, “Status Report on the Third Round of the NIST Post-Quantum Cryptography Standardization Process,” NISTIR 8413, Sep. 2022.

[33] ETSI, “CYBER; Quantum-safe hybrid key exchanges,” ETSI TS 103 744, V1.2.1, Mar. 2025. [Profiles restricted to ML-KEM-512/768/1024 combined with NIST P-256/384, Brainpool256r1/384r1, X25519, or X448.]

[34] European Supervisory Authorities, “Designation of Critical ICT Third-Party Providers under DORA,” 18 Nov. 2025. [19 CTPPs designated including AWS, Google Cloud, Microsoft Ireland, IBM, Oracle, SAP, Deutsche Telekom, Orange, Equinix, Bloomberg, LSEG, TCS, Euroclear, Clearstream, SWIFT.]

[35] OpenEuroLLM, consortium launch, 1 Feb. 2025. [20-member consortium, €37.4M budget (€20.6M from Digital Europe Programme). No dedicated security model stream.]

[36] EuroHPC JU, “AI Factories programme,” 2024–2026. [19 AI Factories across 16 Member States plus 13 Antennas; cumulative EU + Member State funding exceeds €2.7B. Cybersecurity listed as a vertical at IT4LIA (CINECA), PIAT (Poznań), MeluXina-AI (Luxembourg).]

[37] ETSI, “Zero-touch network and Service Management (ZSM); Closed-Loop Automation,” ETSI GS ZSM 009, V1.1.1, Jun. 2021.

[38] GSMA, “Open Gateway programme,” MWC 2026. [86 operator groups, 300+ networks, 80% of global mobile connections.]

[39] European Commission, “IPCEI-CIS,” approved 5 Dec. 2023. [€1.2B public + €1.4B private (€2.6B total), 19 direct participants, 12 Member States. ICRA Reference Architecture v2.0 published January 2026.]

[40] European Parliament and Council, “Regulation (EU) 2025/327 on the European Health Data Space (EHDS),” *Official Journal of the European Union*, Mar. 2025. [Entered into force 26 March 2025.]

[41] DINUM, “Bilan 2025 du cloud public,” published 24 Mar. 2026. [Public cloud purchases at €84M (+62% YoY), 70% to European providers market-wide, 99% within State perimeter.]

[42] S. Dustdar, V. Casamayor Pujol, and P. K. Donta, “On distributed computing continuum systems,” *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 4092–4105, Apr. 2023, doi: 10.1109/TKDE.2022.3142856

[43] L. F. Bittencourt et al., “The Internet of Things, fog and cloud continuum: Integration and challenges,” *Internet of Things*, vol. 3–4, pp. 134–155, Oct. 2018, doi: 10.1016/j.iot.2018.09.005

[44] D. Milojevic, “The edge-to-cloud continuum,” *Computer*, vol. 53, no. 11, pp. 16–25, Nov. 2020, doi: 10.1109/MC.2020.3007297

[45] European Alliance for Industrial Data, Edge and Cloud, *European Industrial Technology Roadmap for the Next-Generation Cloud-Edge*, European Commission, DG CONNECT, Brussels, Belgium, 2024. [Online]. Available: <https://digital-strategy.ec.europa.eu/en/policies/cloud-alliance>

[46] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, “Edge computing: Vision and challenges,” *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, Oct. 2016, doi: 10.1109/JIOT.2016.2579198

[47] M. Satyanarayanan, “The emergence of edge computing,” *Computer*, vol. 50, no. 1, pp. 30–39, Jan. 2017, doi: 10.1109/MC.2017.9

[48] M. Chiang and T. Zhang, “Fog and IoT: An overview of research opportunities,” *IEEE Internet Things J.*, vol. 3, no. 6, pp. 854–864, Dec. 2016, doi: 10.1109/JIOT.2016.2584538

- [49] K. Cao, S. Hu, Y. Shi, A. W. Colombo, S. Karnouskos, and X. Li, "A survey on edge and edge-cloud computing assisted cyber-physical systems," *IEEE Trans. Ind. Inform.*, vol. 17, no. 11, pp. 7806–7819, Nov. 2021, doi: 10.1109/TII.2021.3073066
- [50] ETSI, "Multi-access Edge Computing (MEC); Framework and reference architecture," ETSI GS MEC 003 V3.1.1, Mar. 2022. [Online]. Available: ETSI GS MEC 003 (PDF)
- [51] 6G Infrastructure Association Vision Working Group, *European Vision for the 6G Network Ecosystem*, White Paper, Nov. 2024. [Online]. Available: <https://6g-ia.eu/wp-content/uploads/2024/11/european-vision-for-the-6g-network-ecosystem.pdf>
- [52] National Institute of Standards and Technology, *The NIST Cybersecurity Framework (CSF) 2.0*, NIST Cybersecurity White Paper (CSWP) 29, Gaithersburg, MD, USA, Feb. 26, 2024, doi: 10.6028/NIST.CSWP.29
- [53] A. Alwarafy, K. A. Al-Thelaya, M. Abdallah, J. Schneider, and M. Hamdi, "A survey on security and privacy issues in edge-computing-assisted Internet of Things," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4004–4022, Mar. 15, 2021, doi: 10.1109/JIOT.2020.3015432.
- [54] European Parliament and Council of the European Union, "Regulation (EU) 2019/881 of 17 April 2019 on ENISA and on information and communications technology cybersecurity certification (Cybersecurity Act)," *Off. J. Eur. Union*, L 151, pp. 15–69, Jun. 7, 2019. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2019/881/oj>
- [55] European Union Agency for Cybersecurity (ENISA), *ENISA Threat Landscape 2025: July 2024 to January 2026*, Heraklion, Greece, January 2026, doi: 10.2824/2445233
- [56] European Union Agency for Cybersecurity (ENISA), *Good Practices for Security of IoT — Secure Software Development Lifecycle*, Heraklion, Greece, Nov. 2019, ISBN 978-92-9204-316-2, doi: 10.2824/742784. [Online]. Available: <https://www.enisa.europa.eu/publications/good-practices-for-security-of-iot-1>
- [57] S. Rose, O. Borchert, S. Mitchell, and S. Connolly, *Zero Trust Architecture*, NIST Special Publication 800-207, Gaithersburg, MD, USA, Aug. 2020, doi: 10.6028/NIST.SP.800-207
- [58] A. Cavoukian, *Privacy by Design: The 7 Foundational Principles*, Information and Privacy Commissioner of Ontario, Toronto, ON, Canada, Aug. 2009, rev. Jan. 2012. DOI: 10.4018/978-1-61350-501-4.ch007
- [59] Commission Implementing Regulation (EU) 2024/482 of 31 January 2024 laying down rules for the application of Regulation (EU) 2019/881 as regards the adoption of the European Common Criteria-based cybersecurity certification scheme (EUCC), *Off. J. Eur. Union*, L 2024/482, Feb. 1, 2024. [Online]. Available: https://eur-lex.europa.eu/eli/reg_impl/2024/482/oj
- [60] T. Madiaga, "Digital sovereignty for Europe," European Parliamentary Research Service (EPRS), Brussels, Belgium, EPRS Ideas Paper PE 651.992, Jul. 2020. [Online]. Available: EPRS Ideas Paper PE 651.992
- [61] H. Farrand Carrapico and B. Farrand, "Cybersecurity trends in the European Union: Regulatory mercantilism and the digitalisation of geopolitics," *JCMS: J. Common Market Studies*, vol. 62, no. S1, pp. 147–158, Jul. 2024, doi: 10.1111/jcms.13654
- [62] L. Floridi, "The fight for digital sovereignty: What it is, and why it matters, especially for the EU," *Philosophy & Technology*, vol. 33, no. 3, pp. 369–378, Sep. 2020, doi: 10.1007/s13347-020-00423-6.
- [63] R. Bellanova, H. Carrapico, and D. Duez, "Digital/sovereignty and European security integration: An introduction," *European Security*, vol. 31, no. 3, pp. 337–355, 2022, doi: 10.1080/09662839.2022.2101887
- [64] R. Roman, J. Lopez, and M. Mambo, "Mobile edge computing, Fog et al.: A survey and analysis of security threats and challenges," *Future Gener. Comput. Syst.*, vol. 78, pt. 2, pp. 680–698, Jan. 2018, doi: 10.1016/j.future.2016.11.009.
- [65] D. Ranaweera, A. D. Jurcut, and M. Liyanage, "Survey on multi-access edge computing security and privacy," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 2, pp. 1078–1124, 2nd Quart. 2021, doi: 10.1109/COMST.2021.3062546
- [66] J. Linghe Kong, Jinlin Tan, Junqin Huang, Guihai Chen, Shuaitian Wang, Xi Jin, Peng Zeng, Muhammad Khan, and Sajal K. Das "Edge-computing-driven Internet of Things: A survey," *ACM Comput. Surv.*, vol. 55, no. 8, Dec. 2022, Art. no. 174, doi: 10.1145/3555308
- [67] R. Chandramouli and Z. Butcher, *A Zero Trust Architecture Model for Access Control in Cloud-Native Applications in Multi-Cloud Environments*, NIST SP 800-207A, Gaithersburg, MD, USA, Sep. 2023, doi: 10.6028/NIST.SP.800-207A [Online] <https://csrc.nist.gov/pubs/sp/800/207/a/final>